

Chapter

Histogram And Ogive

1. The following distribution gives the daily wages of 60 workers of a factory.

Daily income in ₹	Number of workers (f)
200–300	6
300–400	10
400–500	14
500–600	16
600–700	10
700–800	4

Main 2023

Use graph paper to answer this question. Take 2 cm = ₹100 along one axis and 2 cm = 2 workers along the other axis. Draw a histogram and hence find the mode of the given distribution.

2. Monthly Daily wages of 110 workers, obtained in a survey, are tabulated below :

Wages (in ₹)	Number of people
6000-7000	8
7000-8000	36
8000-9000	18
9000-10000	15
10000-11000	11
11000-12000	8
12000-13000	4

Use graph paper to answer this question. Draw a histogram and hence find the mode of the given distribution.

3. Use graph paper to answer this question. Draw a histogram and hence estimate the mode for the following frequency distribution.

Class	0-20	20-40	40-60	60-80	80-100	100-120
Frequency	7	16	20	10	5	6

4. Marks obtained by 100 students in an examination are given below :

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Number of Students	5	15	20	28	20	12

Draw a histogram for the given data using a graph paper and find the mode.
Main 2022

5. The following distribution gives the distribution of life times of washing machines of a certain company : SQP 2022

Life time (in hours)	No. of washing machines
1000-1200	15
1200-1400	60
1400-1600	68
1600-1800	86
1800-2000	75
2000-2200	61
2200-2400	45

Draw a histogram for the above data using a graph paper and locate the mode.

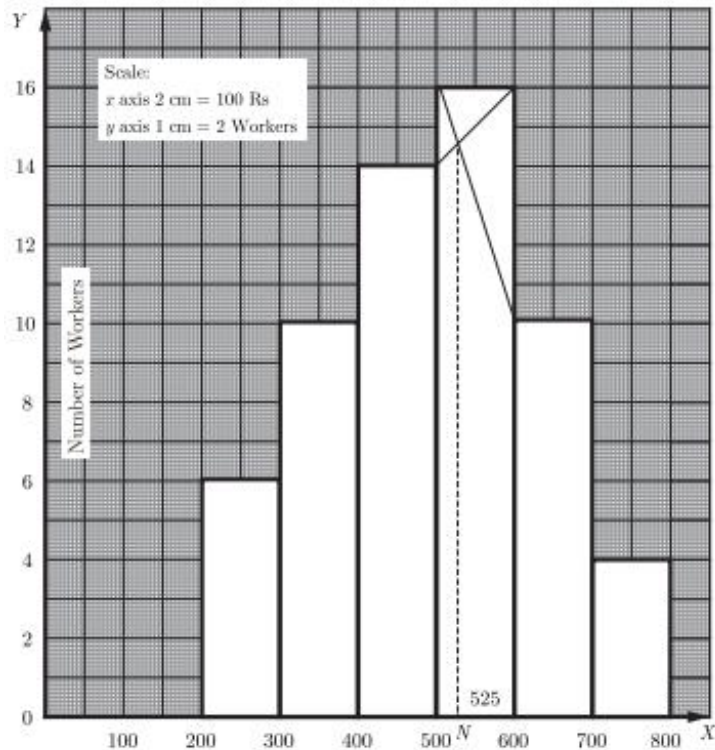
6. Use a graph paper for this question, the daily pocket expenses of 100 students in a school are given below : SQP 2021

Pocket expenses (in Rs.)	Number of students
0-10	7
10-20	5
20-30	14
30-40	21
40-50	25
50-60	15
60-70	6
70-80	7

Draw a histogram representing the above distribution and estimate the mode from the graph.

Solution

1. The histogram from the given frequency table is shown below.



In the highest rectangle draw two straight lines AC and BD, which intersect at point P. Through point P, draw a vertical line to meet the x-axis at N. The abscissa of the point represents 525. Hence, the required mode is 525.

2.

Step 1: Draw the Histogram

- Take **wages (₹)** on the horizontal axis.
- Take **number of workers** on the vertical axis.
- Draw rectangles for each class interval.
- The tallest rectangle corresponds to the class 7000–8000, showing the highest frequency.

Step 2: Identify the Modal Class

The class with the highest frequency is:

$$7000-8000$$

This is the **modal class**.

Step 3: Calculate the Mode

Use the formula for mode of grouped data:

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Where:

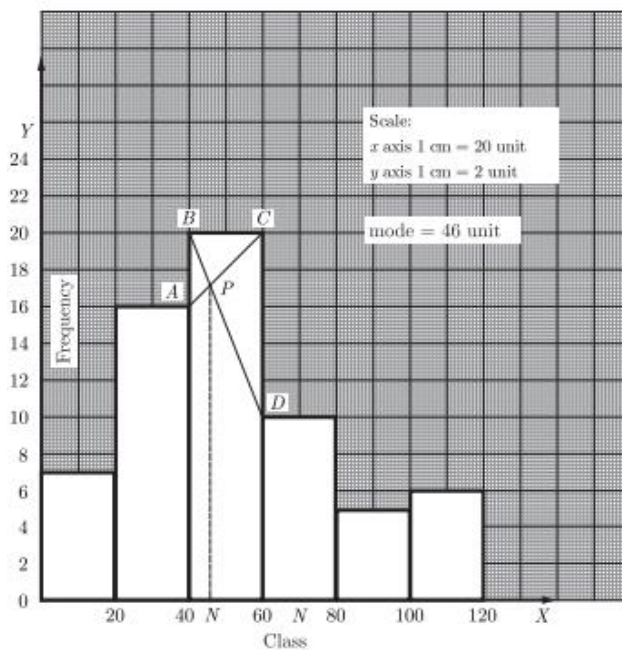
- $l = 7000$ (lower limit of modal class)
- $h = 1000$ (class width)
- $f_1 = 36$ (frequency of modal class)
- $f_0 = 8$ (preceding class frequency)
- $f_2 = 18$ (succeeding class frequency)

$$\begin{aligned}\text{Mode} &= 7000 + \frac{36 - 8}{2(36) - 8 - 18} \times 1000 \\ &= 7000 + \frac{28}{46} \times 1000 \\ &\approx 7000 + 609 \\ &\approx 7609\end{aligned}$$

Answer:

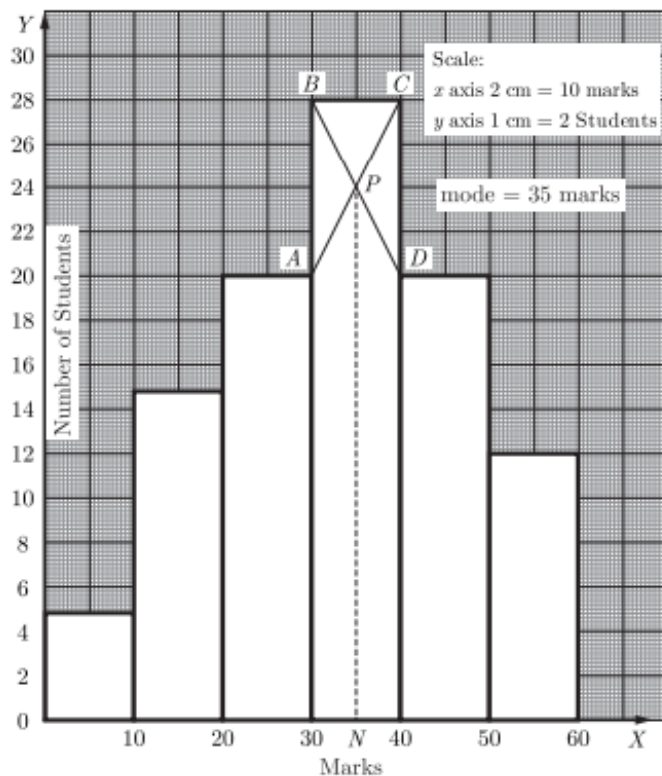
$$\text{Mode} \approx ₹7600$$

3. The histogram from the given frequency table is shown below.



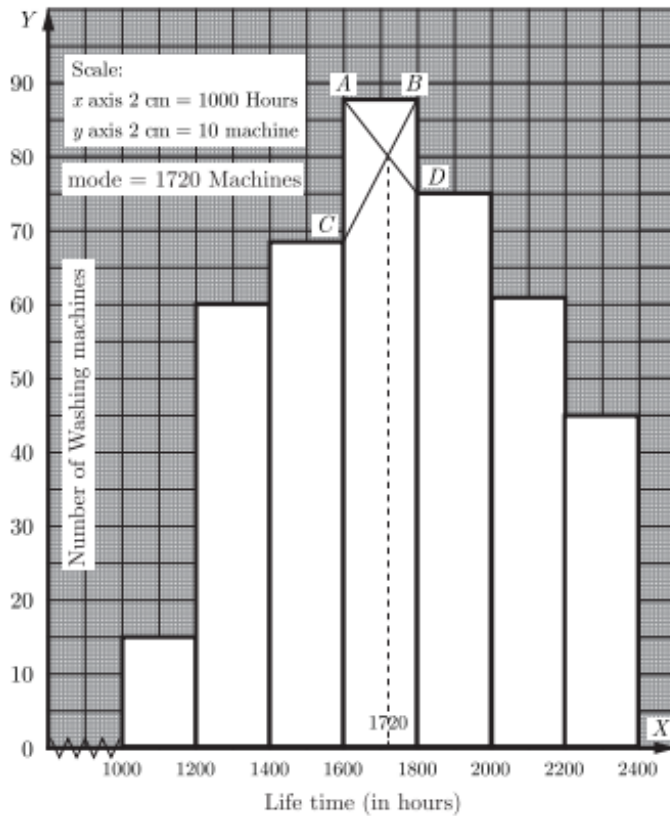
In the highest rectangle draw two straight lines AC and BD, which intersect at point P. Through point P, draw a vertical line to meet the X -axis at N . The abscissa of the point represents 46. Hence, the required mode is 46.

4. The histogram from the given frequency table is shown below.



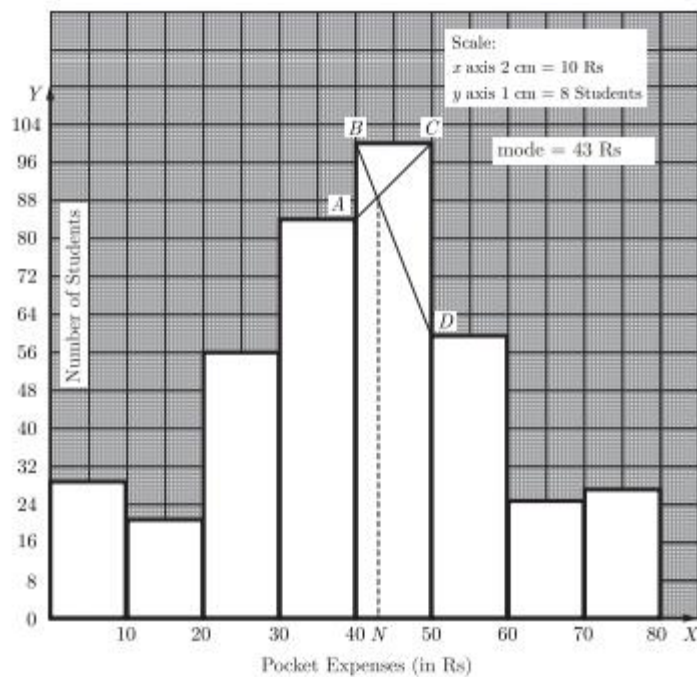
In the highest rectangle draw two straight lines AC and BD, which intersect at point P. Through point P, draw a vertical line to meet the x-axis at N. The abscissa of the point represents 35. Hence, the required mode is 35.

5. The histogram from the given frequency table is shown below.



In the highest rectangle, draw two straight lines AC and BD (as shown in figure) which intersect at P. Through point P, draw a vertical line to meet the x-axis at N. The abscissa of the point represent 17220. Hence, the required mode is 1720 hours.

6. The histogram from the given frequency table is shown below.



In the highest rectangle draw two straight lines AC and BD, which intersect at point P. Through point P, draw a vertical line to meet the x -axis at N. The abscissa of the point represents 43. Hence, the required mode is Rs 43.